

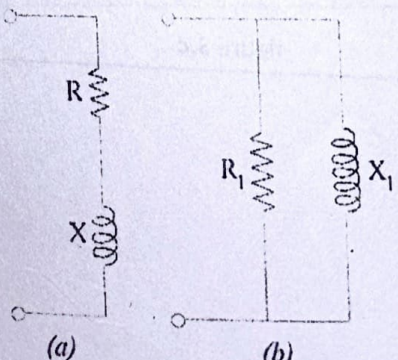
(Please write your Enrollment Number)

Enrollment No. \_\_\_\_\_

**MID-TERM EXAMINATION**  
(Course Name: B. Tech. (Semester: II))  
(March 2025) OFF LINE mode

|                       |                                                           |
|-----------------------|-----------------------------------------------------------|
| Subject Code: BEC 101 | Subject: Basics of Electrical and Electronics Engineering |
| Time : 1 ½ Hours      | Maximum Marks :30                                         |

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

|    |                                                                                                                                                                                                                                                                                                                                                                                                                        |            |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q1 | Attempt all parts.                                                                                                                                                                                                                                                                                                                                                                                                     | (2.5*4=10) |
|    | a) Define Active, Reactive and Apparent Power. Also show them on the Power Triangle.                                                                                                                                                                                                                                                                                                                                   |            |
|    | b) Explain ideal voltage source and real voltage source.                                                                                                                                                                                                                                                                                                                                                               |            |
|    | c) Explain the concept of Resonance in series RLC circuit. Also draw the Resonance curve.                                                                                                                                                                                                                                                                                                                              |            |
|    | d) A coil having an inductance of 50 mH and resistance 10 $\Omega$ is connected in series with a 25 $\mu$ F capacitor across a 200 V ac supply. Calculate<br>(1) resonance frequency of the Circuit<br>(2) current flowing at resonance<br>(3) value of Quality Factor by using different data.                                                                                                                        |            |
| Q2 | Attempt any two parts.                                                                                                                                                                                                                                                                                                                                                                                                 | (5*2=10)   |
|    | a) State and prove Maximum Power Theorem.                                                                                                                                                                                                                                                                                                                                                                              |            |
|    | b) In a series circuit containing pure resistance and a pure inductance, the current and the voltage are expressed as:<br>$I(t) = 5 \sin(314t + 2\pi/3)$ and $V(t) = 15 \sin(314t + 5\pi/6)$<br>(1) What is the impedance of the circuit?<br>(2) What is the value of the resistance?<br>(3) What is the inductance in henrys?<br>(4) What is the average power drawn by the circuit?<br>(5) What is the power factor? |            |
|    | c) The series and parallel circuits shown in Figure 2.c, have the same impedance and the same power factor. If $R = 3 \Omega$ and $X = 4 \Omega$ , find the values of $R_1$ and $X_1$ . Also, find the impedance and power factor.                                                                                                                                                                                     |            |
|    |  <p>Figure 2.c</p>                                                                                                                                                                                                                                                                                                                  |            |

Q3

Attempt any two parts

(5\*2=10)

a) Find Thevenin equivalent circuit for the network shown in Figure. 3.a

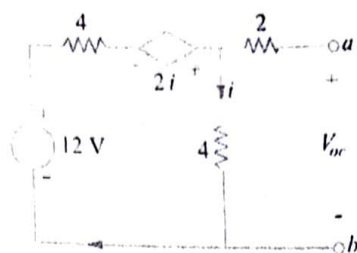


Figure. 2.a

b) Using Superposition theorem, find the value of the output voltage of the circuit as shown in figure 3.b

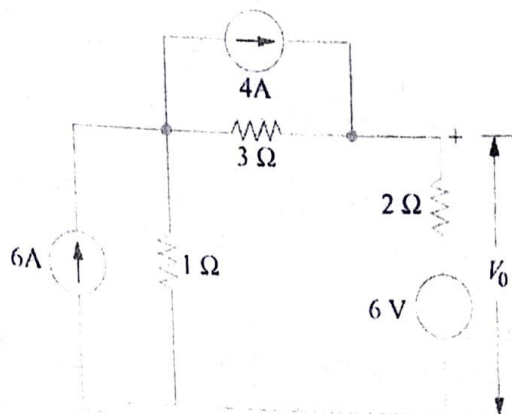


figure 3.b

c) Using Node voltage method, find the current in the 3Ω resistance for the network as shown in figure 3.c

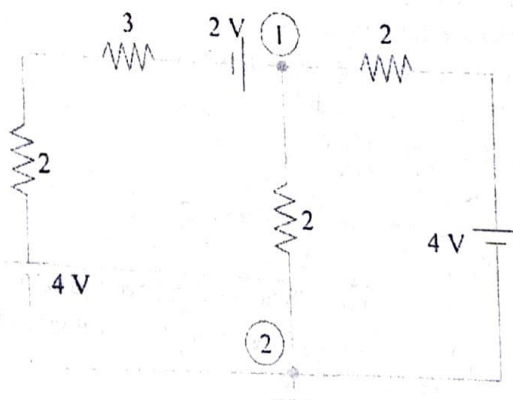


figure 3.c